

OPERATING SYSTEMS AND SYSTEM PROGRAMMING(OSSP)

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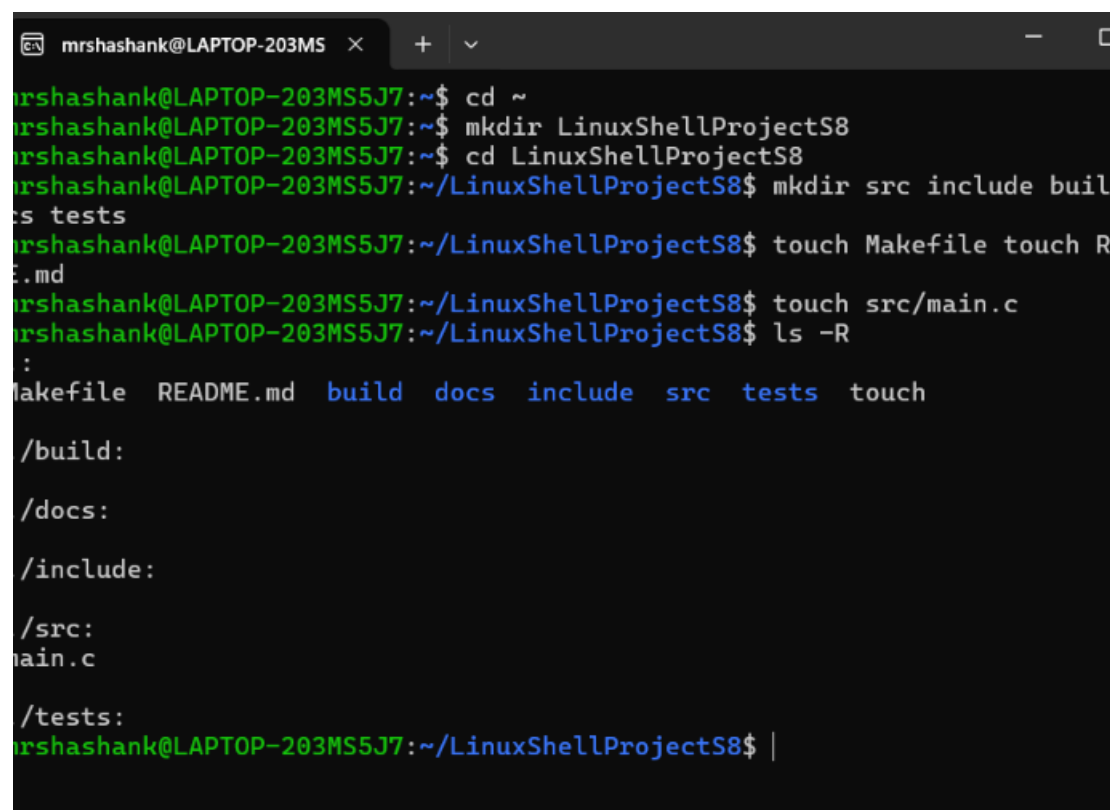
ROLL NO: 2520030215

SEC: 08

SKILL -2

1. **AIM:**To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction



```
mrshashank@LAPTOP-203MS5J7:~$ cd ~
mrshashank@LAPTOP-203MS5J7:~$ mkdir LinuxShellProjectS8
mrshashank@LAPTOP-203MS5J7:~$ cd LinuxShellProjectS8
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ mkdir src include build docs tests
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ touch Makefile touch README.md
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ touch src/main.c
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ ls -R
.:
Makefile  README.md  build  docs  include  src  tests  touch

/build:

/docs:

/include:

/src:
main.c

/tests:
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ |
```

```
mrshashank@LAPTOP-203MS5J7:~$ cd ~
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.:
Makefile  README.md  build  docs  include  src  tests  touch

./build:

./docs:

./include:

./src:
main.c

./tests:
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ nano Makefile
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ nano src/main.c
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ make gcc -Wall src/main.c -o shell
Makefile:4: *** missing separator.  Stop.
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ char input[100];
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'chat' from deb ppp (2.5.2-1+1.2)
  command 'phar' from deb php8.5-cli (8.5.4-0ubuntu1.1)
  command 'phar' from deb php-cli (2:8.5+99ubuntu1)
  command 'pchar' from deb pchar (1.5-7)
  command 'shar' from deb sharutils (1:4.15.2-13)
  command 'chr' from deb chr (0.1.80-1)
  command 'chr' from deb chr-tiny (0.1.80-1)
  command 'car' from deb ucommon-utils (7.0.1-1)
See 'snap info <snapname>' for additional versions.
mrshashank@LAPTOP-203MS5J7:~/LinuxShellProjectS8$ |
```

CODE:

```
</> Makefile

CC = gcc
CFLAGS = -Wall
TARGET = shell

all:
    gcc -Wall src/main.c -o shell

clean:
    rm -f shell
```

```
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        printf("myshell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        if (strcmp(input, "exit\n") == 0)
        {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s", input);
    }

    return 0;
}
```